GAIRA V7 — Phase 04.5

Hierarchical NMF over Frozen CSM Activations

Figure report — all 14 figures with captions

Status	COMPLETE
VERDICT	DISCARD
Selected	plain, K = 3
Meta fingerprint	d24fbc288fe99ada2be47f667c3edf5b
Frozen inputs	atlas / LSM / CSM / theme — all verified, none refitted
CSM class top-1	0.856
Meta class top-1	0.392
CSM macro F1	0.810
Meta macro F1	0.196
Information retained vs CSM	0.185 (floor 0.50)
Class retrieval ratio	0.458 (floor 0.50)
Informativeness floor	FAIL
Axes improved	3 of 8 — all stability, none informative
Robustness delta vs CSM	-0.4123 AURC
Clean accuracy cost	-0.4466 top-1
Completed	2026-08-06 18:34:06 UTC

Committed sources of record: reports/PHASE_04_5_REPORT.md and reports/PHASE_04_5_SCIENTIFIC_AUDIT.md
This is a NEGATIVE result. The CSM layer remains the canonical inference representation.
Figures are embedded from the committed PNGs, so this PDF cannot drift from the run that produced them.

Figures

01 architecture
02 workflow
03 pareto frontier
04 meta heatmap
05 composition
06 activation maps
07 representation comparison
08 robustness curves
09 robustness auc
10 k diagnostic
11 bootstrap stability
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13 geometry overlay
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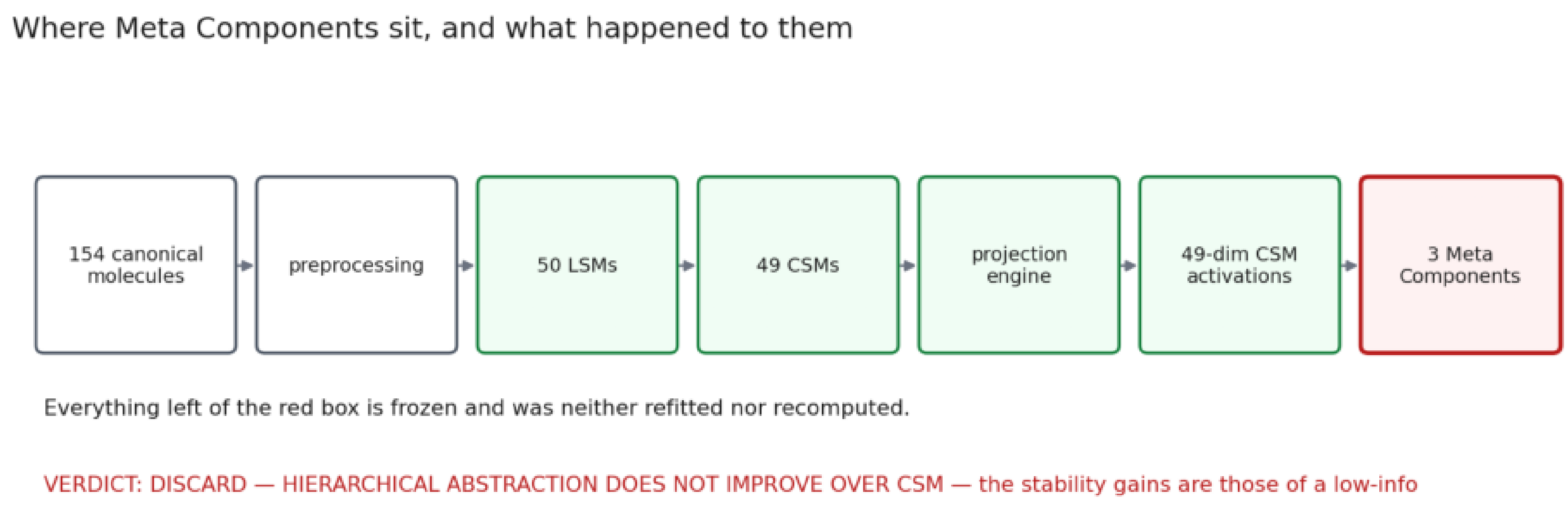
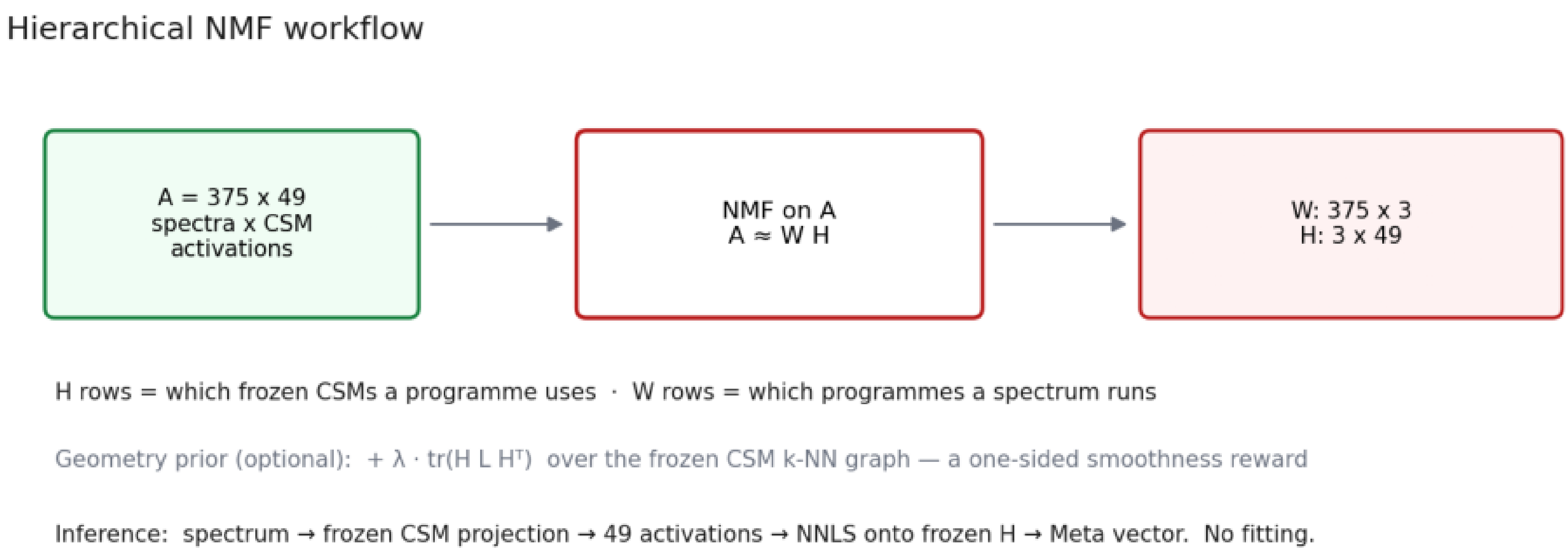
Figure 01



Where Meta Components sit in the architecture, and the verdict. Everything left of the red box is frozen and was neither refitted nor recomputed.

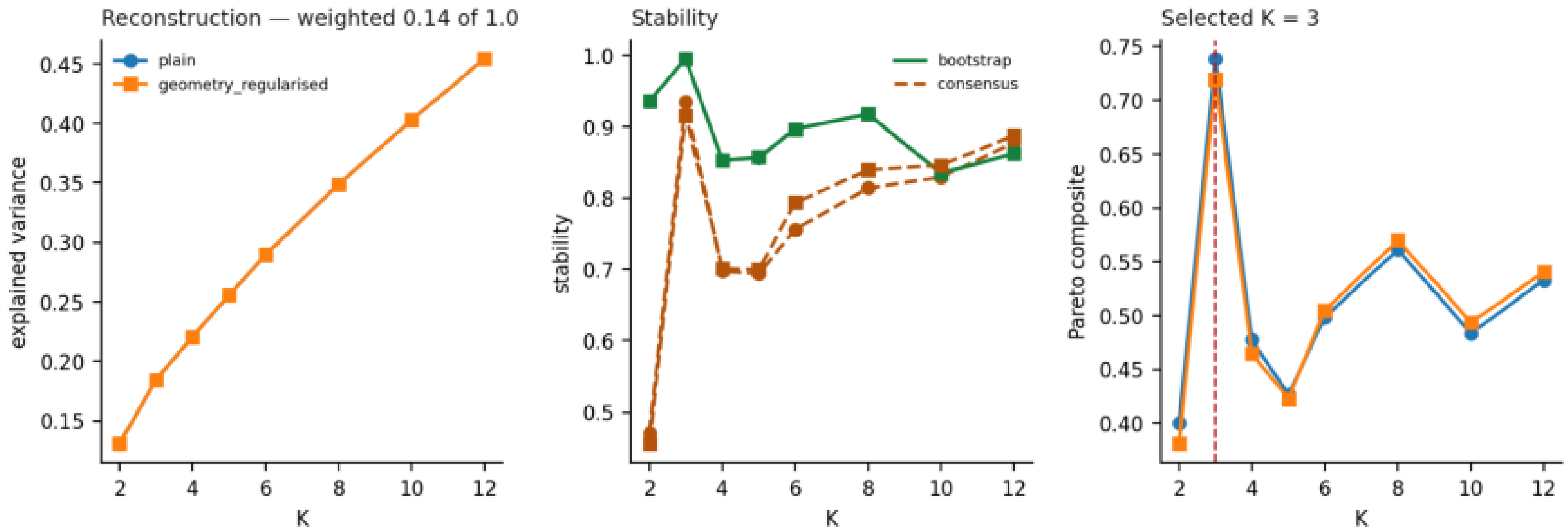
Figure 02



The hierarchical NMF workflow. A is spectra x frozen CSM activations — not spectra, not a similarity matrix, not a graph. The geometry prior is a one-sided smoothness reward that cannot push distant CSMs apart.

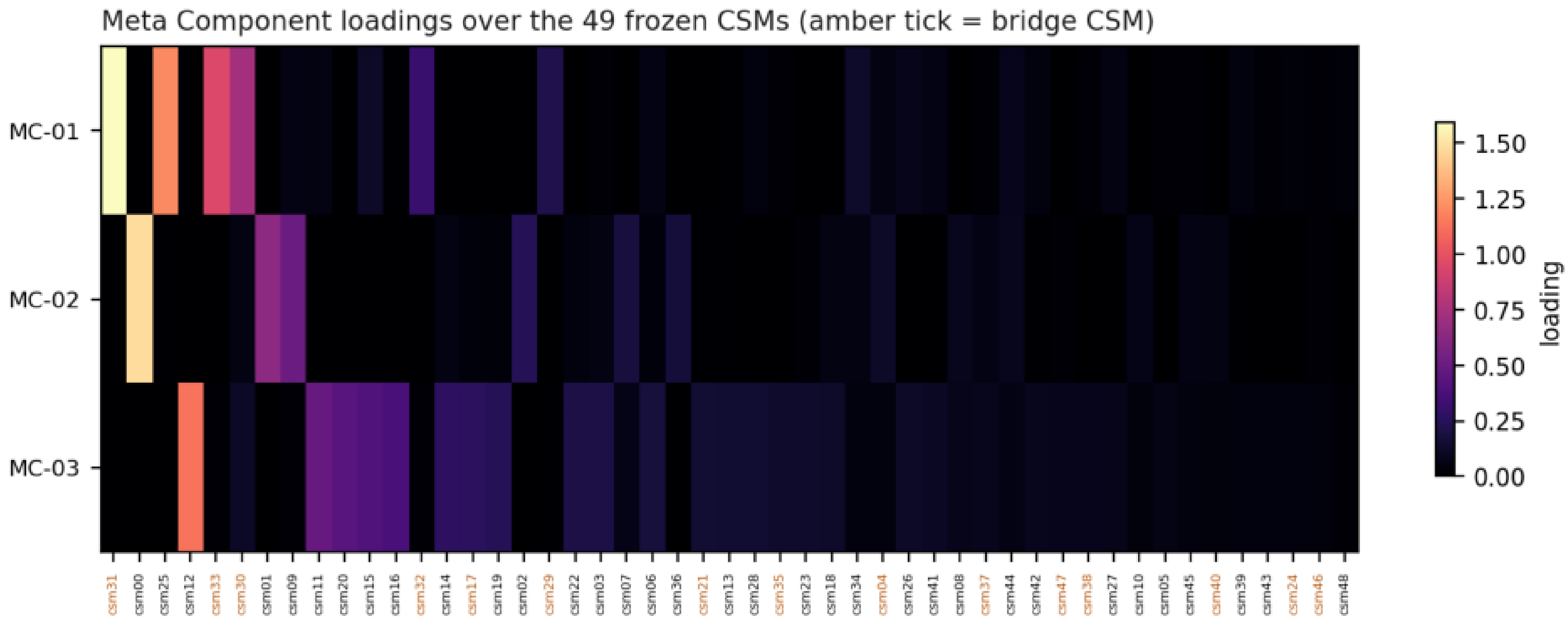
Figure 03

Model selection over eight K and two variants. Stability carries 0.40 of the composite and reconstruction 0.14 —
which is what pulls K down to 3.



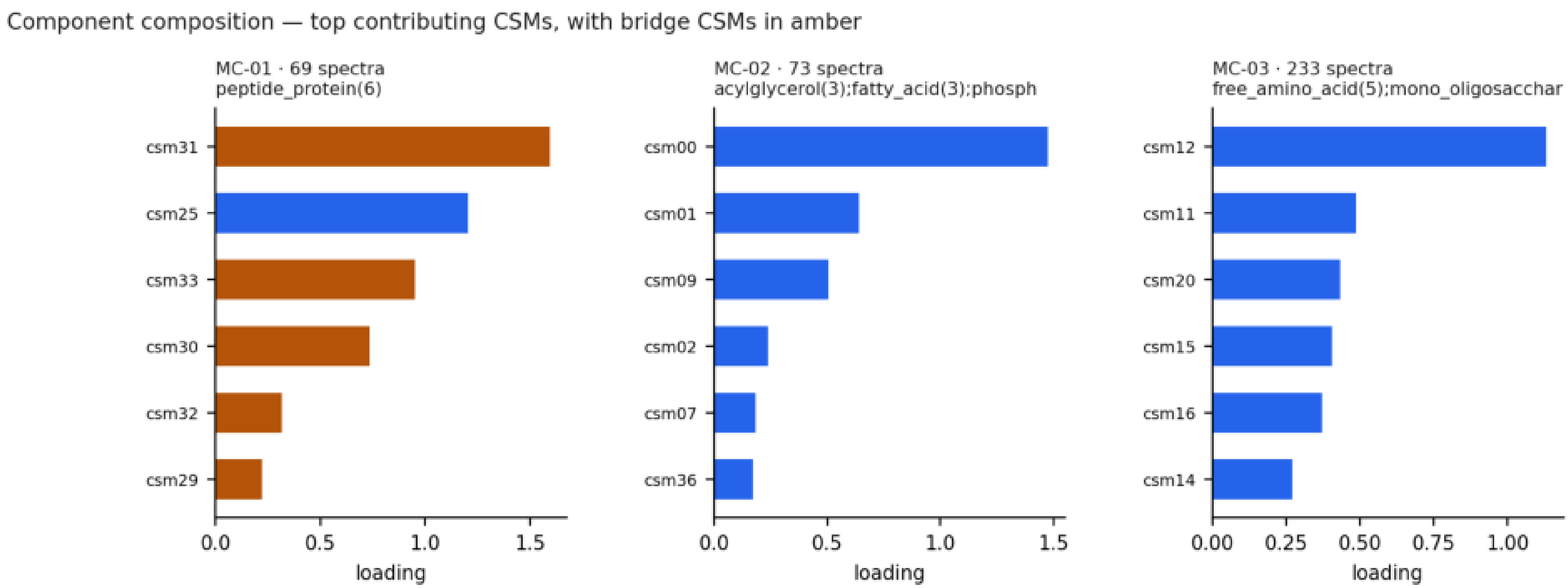
Model selection over eight K and two variants. Stability carries 0.40 of the Pareto composite and reconstruction 0.14, as the brief requires — which is what pulls K down to 3, and K = 3 is where the information has gone.

Figure 04



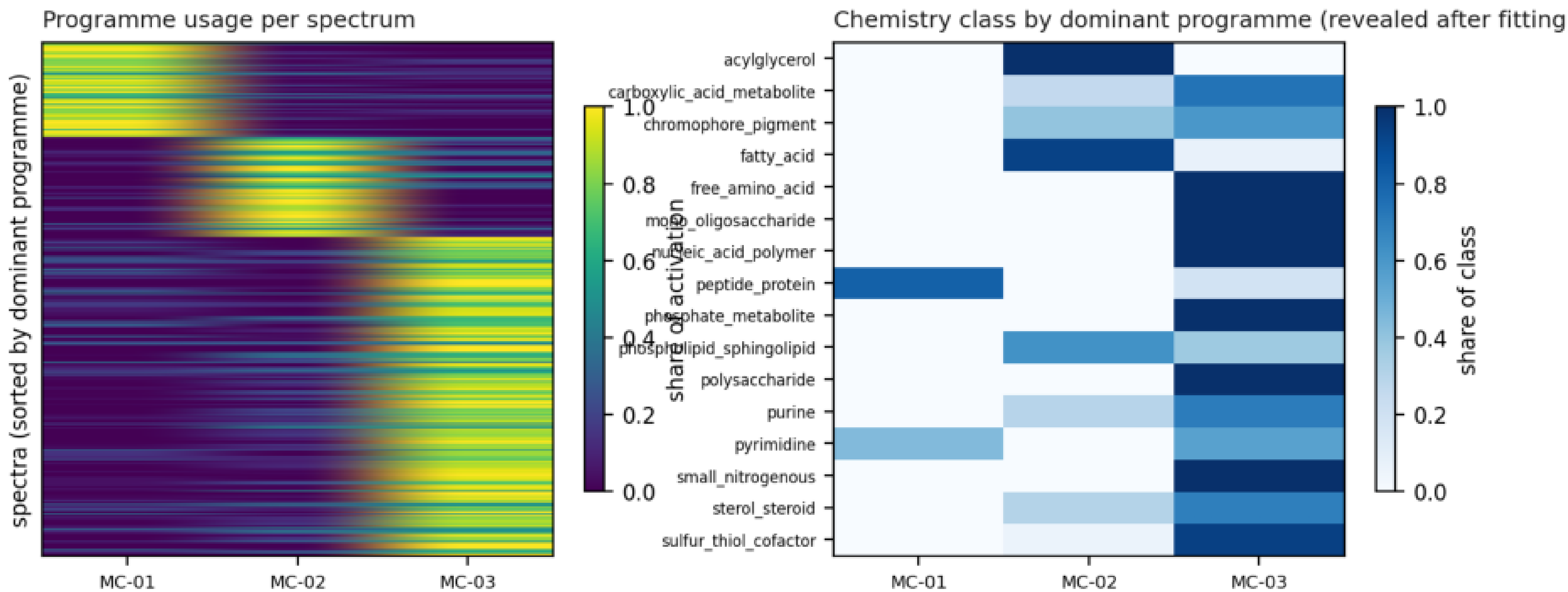
Meta Component loadings over the 49 frozen CSMs. Amber ticks mark Phase 02.5 bridge CSMs; MC-01 loads almost entirely on them.

Figure 05



Component composition. MC-02 is the only component with a clean non-bridge reading — acylglycerol, fatty acid and phospholipid CSMs co-activating, the lipid programme every previous phase has also found.

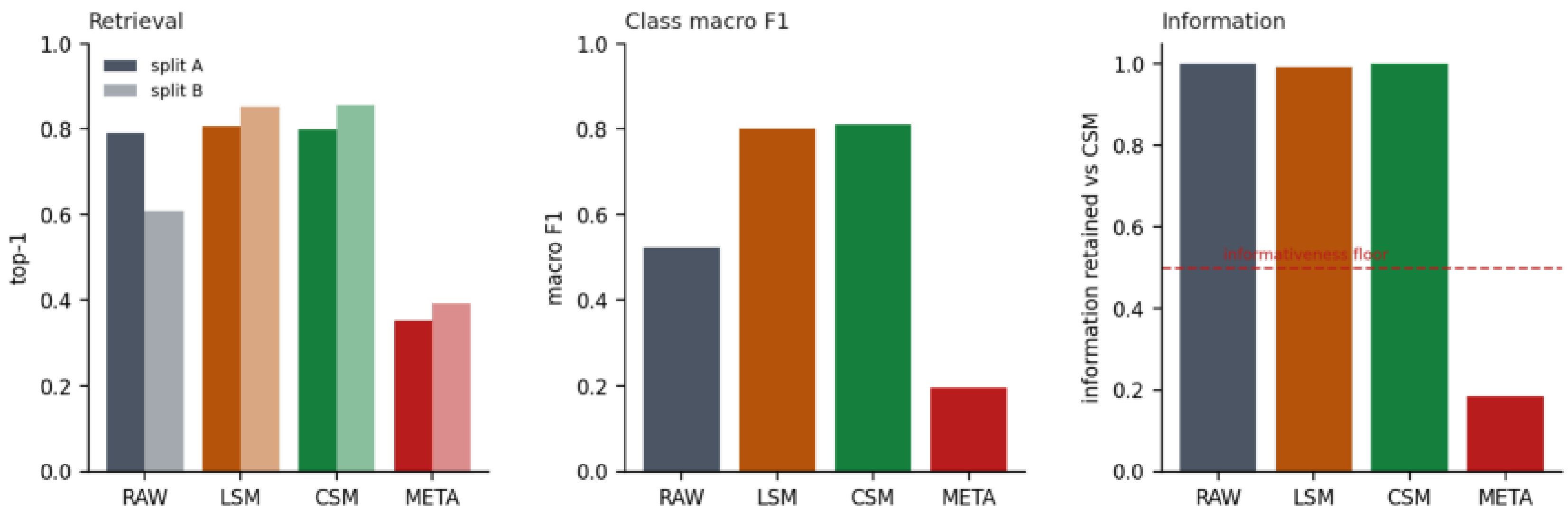
Figure 06



Programme usage per spectrum, and chemistry class by dominant programme. MC-03 dominates 233 of 375 spectra: that is a background, not a programme.

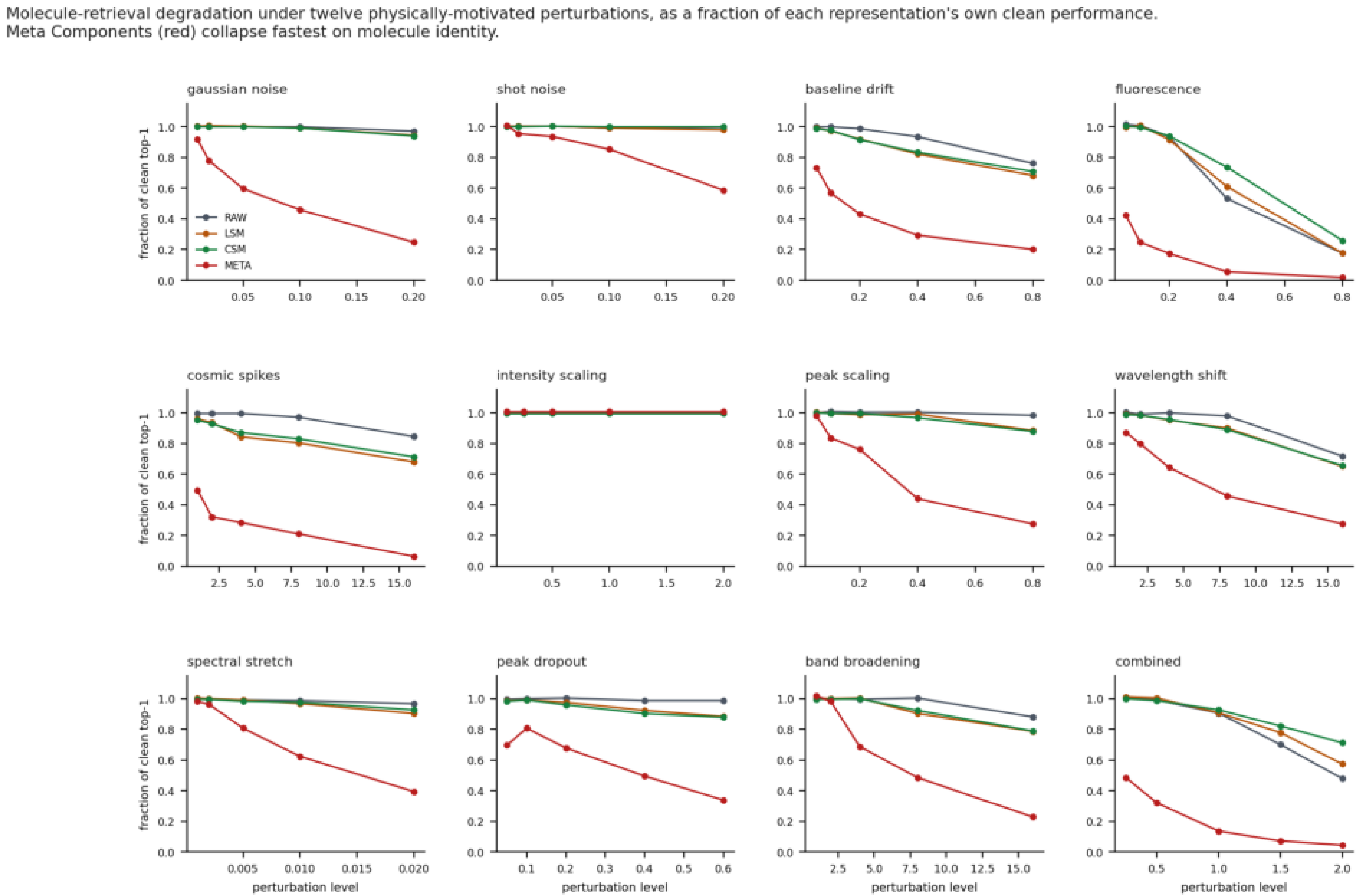
Figure 07

Four representations, identical frozen splits. Meta Components retain 0.185 of the CSM layer's information and 0.458 of its class retrieval.



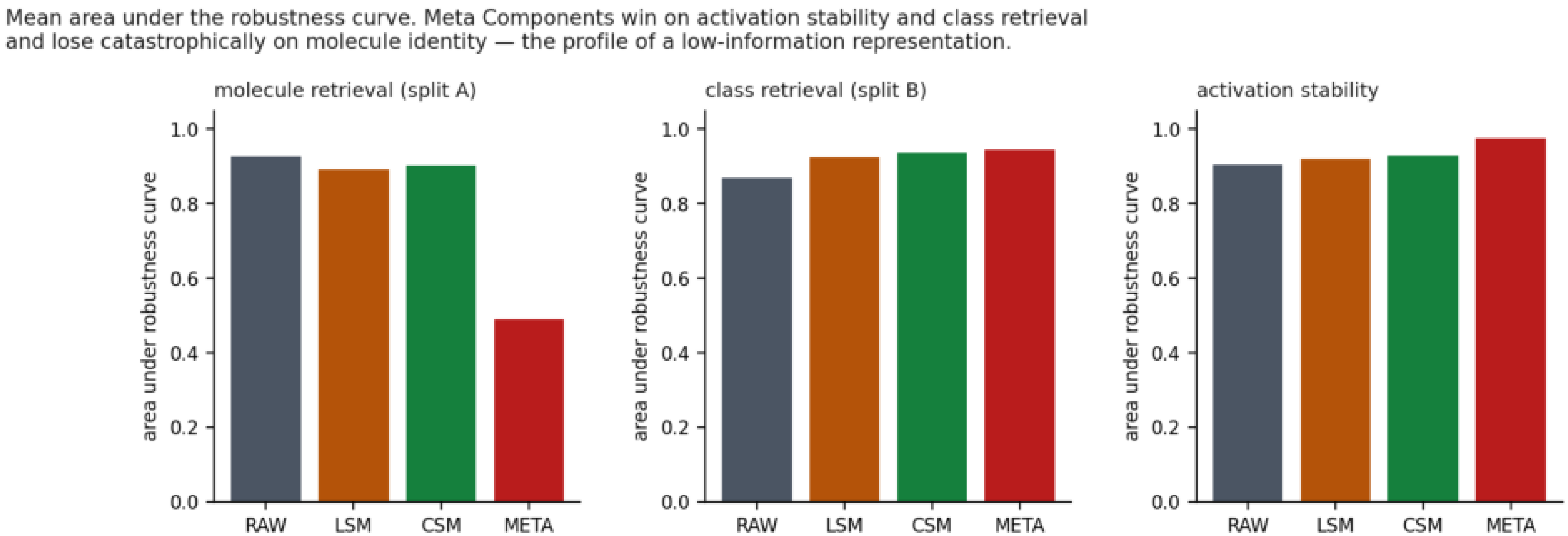
Four representations on identical frozen splits. Meta Components retain 0.185 of the CSM layer's information and 0.458 of its class retrieval, well below the 0.50 informativeness floor.

Figure 08



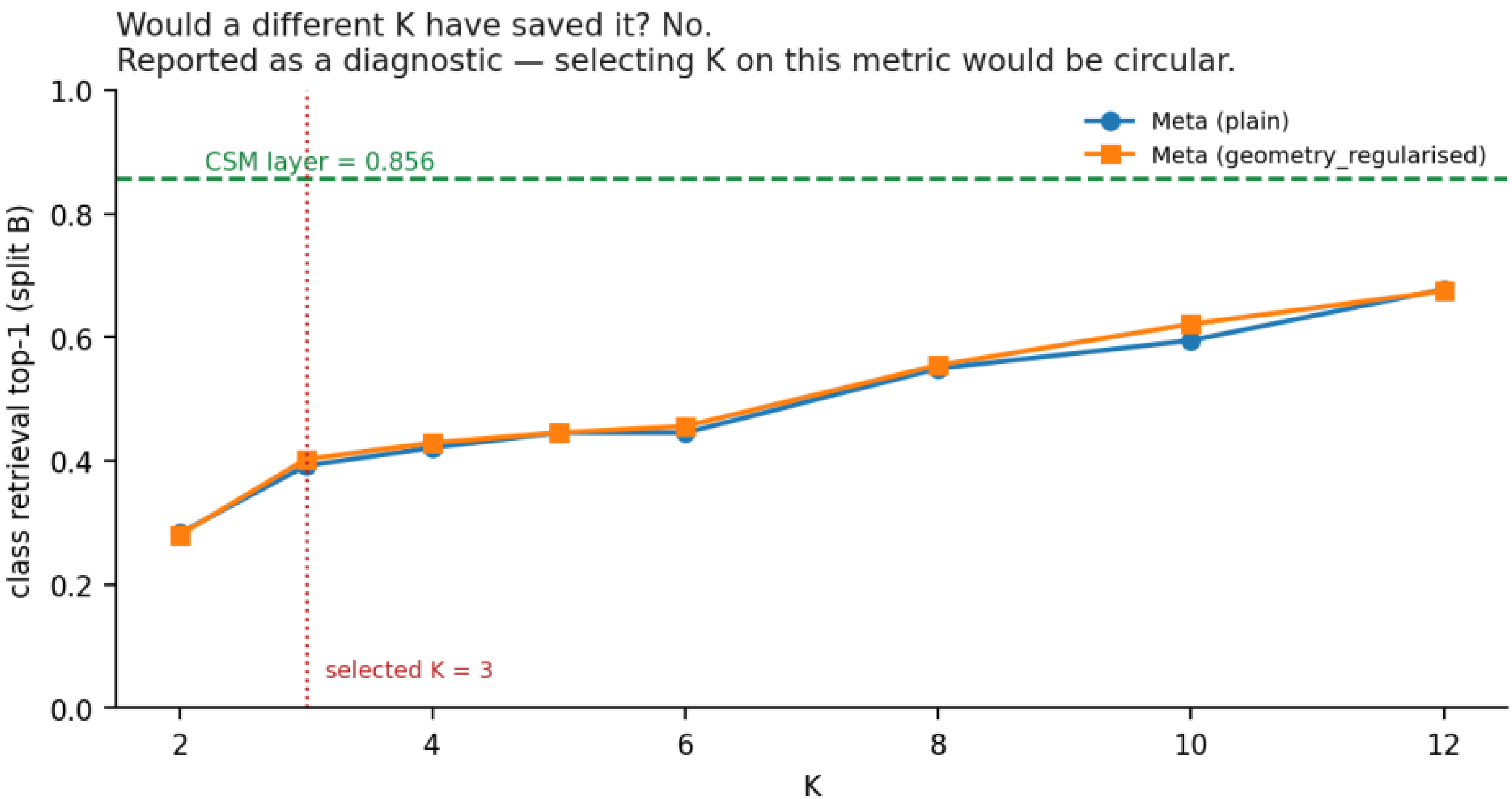
Molecule-retrieval degradation under twelve physically-motivated perturbations, each normalised by that representation's own clean performance. Meta Components collapse fastest on molecule identity under almost every corruption.

Figure 09



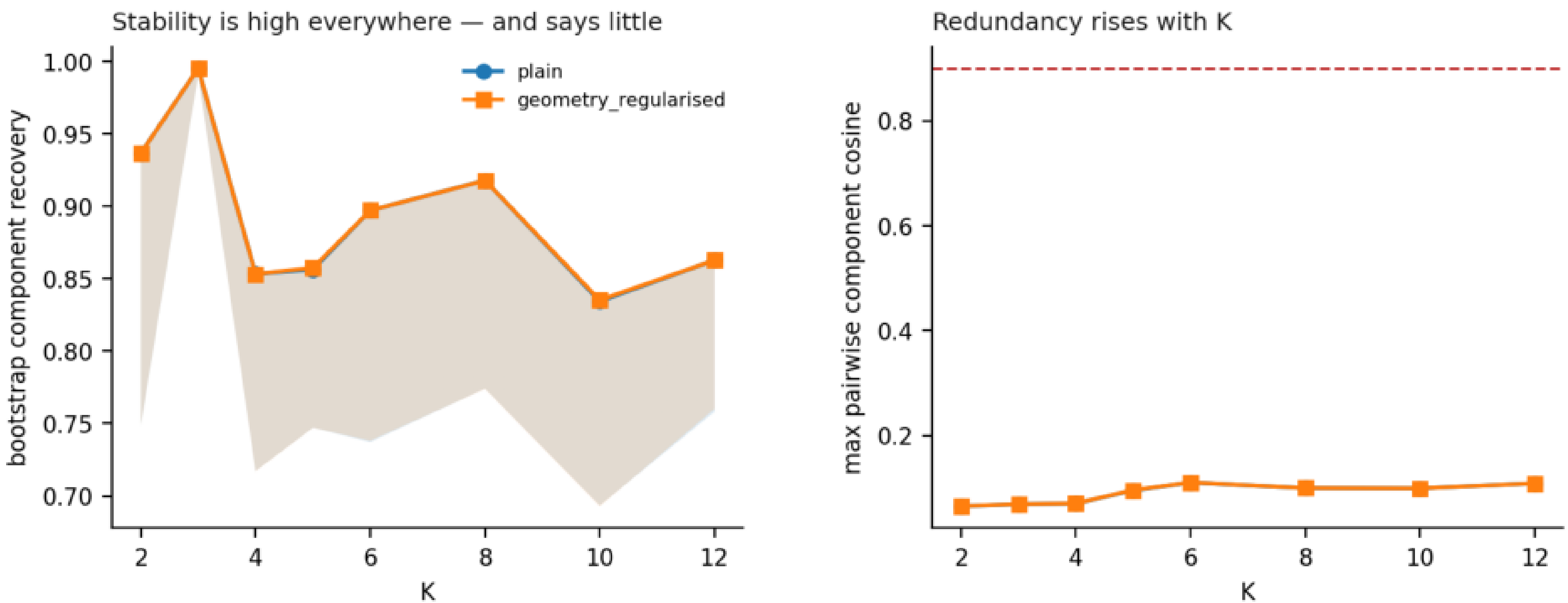
Mean area under the robustness curve. Meta Components win on activation stability and class retrieval and lose catastrophically on molecule identity — the profile of a low-information representation, not of a better one.

Figure 10



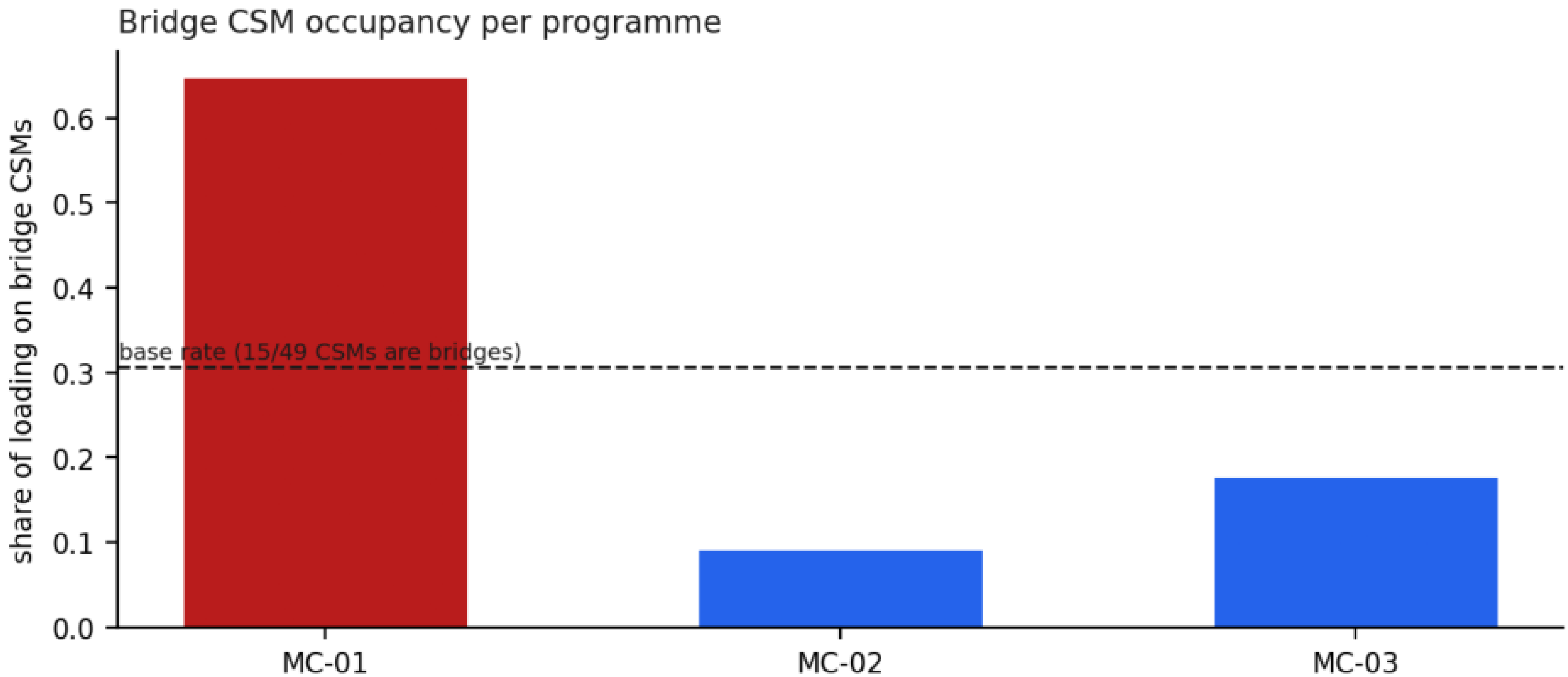
Would a different K have saved it? No. The best achievable class retrieval over all sixteen (variant, K) combinations is 0.677 against the CSM layer's 0.856. Reported as a diagnostic — selecting K on this metric would be circular.

Figure 11



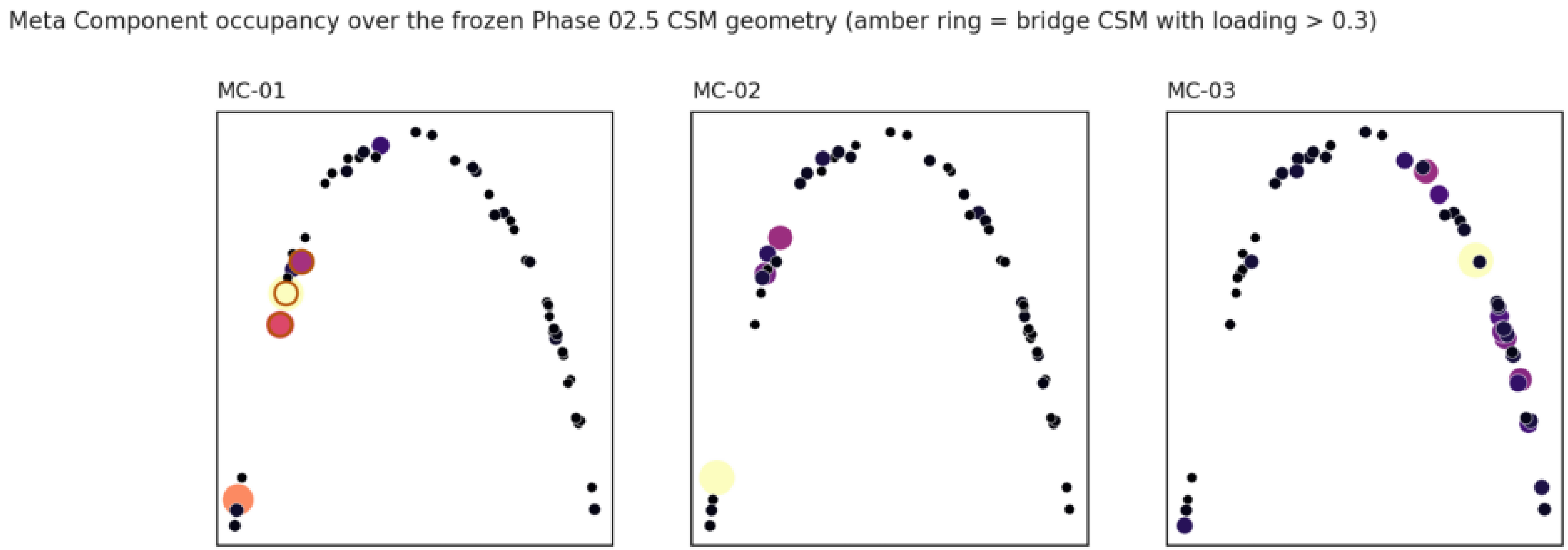
Bootstrap component recovery is high at every K and therefore says very little; component redundancy rises with K.

Figure 12



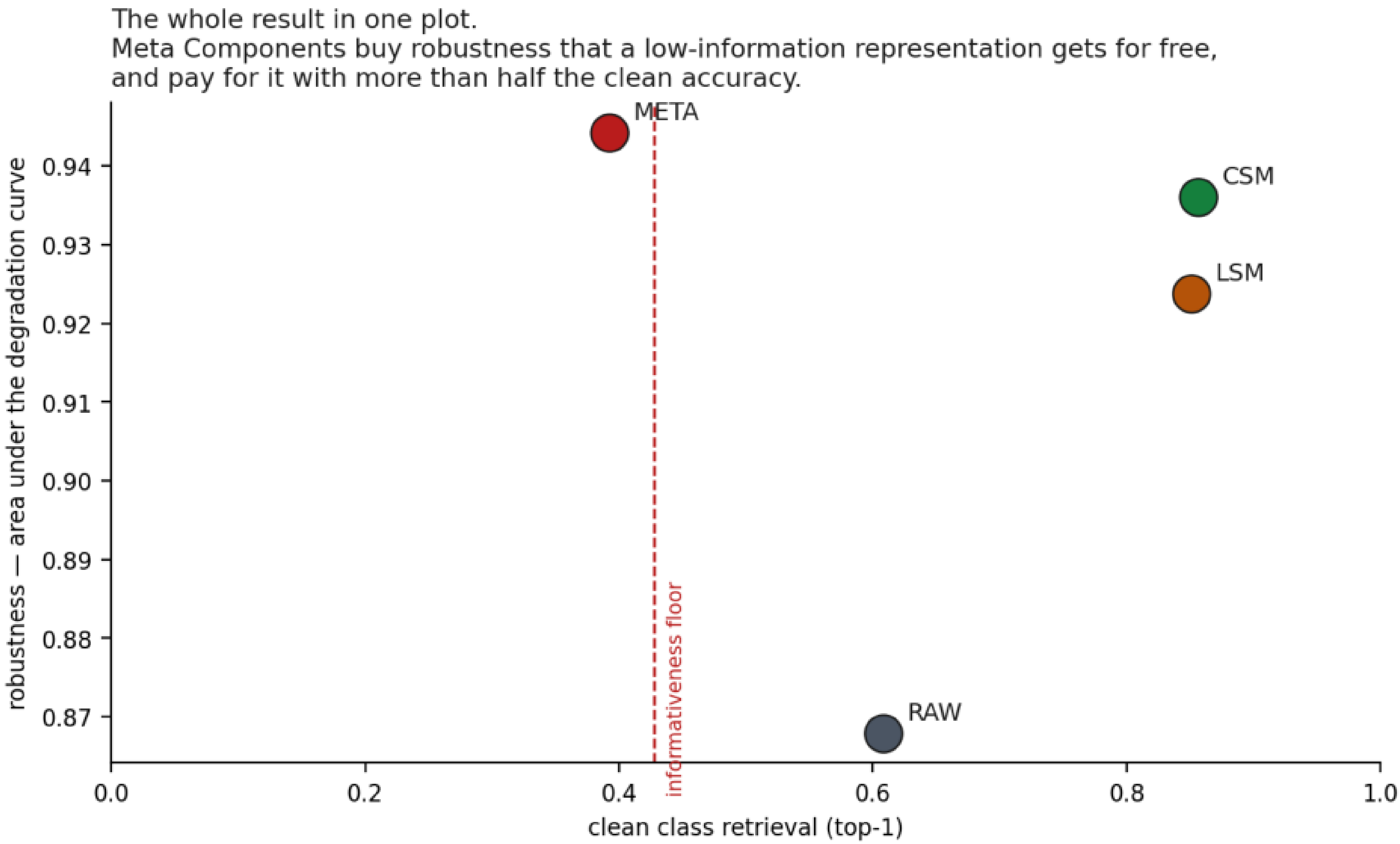
Bridge CSM occupancy per programme against the base rate. MC-01 sits far above it — the one component with a clean single-class signature is built from the CSMs the geometry already flagged as ambiguous.

Figure 13



Meta Component occupancy over the frozen Phase 02.5 CSM geometry.

Figure 14



The whole result in one plot. Meta Components buy robustness that a low-information representation gets for free, and pay for it with more than half the clean accuracy.